

How is the Appalachian Region defined? How many US States are included?

The Appalachian Region is made up of 423 counties across 13 states. The region spans 206,000 square miles and is home to 26.4 million residents. The states included in this region are Alabama, Georgia, Kentucky, Maryland, Mississippi, New York, North Carolina, Ohio, Pennsylvania, South Carolina, Tennessee, Virginia, and West Virginia.

Provide some context on how rural Appalachia compares to the rest of rural America, specifically, in:

- population and demographics: 26.4 million residents across 13 different states
- education: As of 2023, 27.3 percent of residents received a bachelor's degree (Lower than the U.S. percent)
- employment: 95.9 percent employment rate | 4.1 percent unemployment rate
- income and poverty: As of 2023, 14.3 percent of residents live in poverty (Higher than the U.S. percent)

As of 2023, the average income is 55,000 dollars (Lower than the U.S. average)

- broadband access: less than 80 percent of people had broadband subscriptions

Briefly describe each of the five listed goals in the ARC's Strategic Plan for investing in the Appalachian Region

Building Business: Providing assistance and support for small businesses' development

Workforce Ecosystems: Invest in educational programs as well as workforce development programs and address and support physical and mental health of workers and their families

Community Infrastructure: Focus on the availability of basic infrastructure, adopt alternative energy strategies, invest in transportation systems, and invest in business sites and public facilities.

Regional Culture and Tourism: Investment in downtowns, community initiatives to preserve cultural and heritage traditions. Invest in increasing outdoor recreation opportunities.

Leaders and Local Capacity: Empower community leaders, build capacity, and build a shared vision for the growth and development of local communities.

Definition of some variables:

(three-year) average unemployment rate per capita: it is the measure of long term structural unemployment

Market income: is calculated by dividing total personal income by population

Poverty rate: is calculated by dividing the number of persons living below the poverty threshold by the number of persons for whom poverty status has been determined

Mean, Max, Min, and Standard Deviation of poverty rate across the csv file:

The mean of the the poverty rate is : 14.41. This represents the average rate of poverty in the three year period. 14.41% live in poverty.

The standard deviation of the the poverty rate is : 6.1. This means the poverty rate differs from the mean by 6.1% higher or lower than 14.41. This is considered as a high standard deviation so there are areas with much higher or much lower than the average.

The maximum of the the poverty rate is : 59.0. The highest poverty rate across all counties.

The minimum of the the poverty rate is : 1.2. The lowest poverty rate across all counties.

The use of (thousands = “.”):

Without it, it read the numbers as strings and returned an error so the function ensure they are read as numeric values.

Observed in data tables

The data table shows the differences between counties across the U.S. when it comes to poverty, income, and unemployment. Generally speaking, counties with high poverty rates usually have low per capita incomes (PCI) and higher unemployment. Counties with low poverty tend to have high incomes and low unemployment. The table reflects this, but also shows that there are some exceptions even among counties within the same state.

Investigation: South Dakota Counties (Oglala Lakota vs. Union)

Upon further research, these two counties differ heavily but are in the same state. This is because Oglala Lakota County lies in an Indian reservation. This explains why it shows extreme poverty on our tables. Longstanding marginalization and discrimination against indigenous people contribute to the very high poverty rate. Union County is also in South Dakota, but it shows a

high PCI. This is due to its location within the state. It is located near larger economic centers, which explains why its PCI is so high. Overall, it is clear that counties contrast despite being from the same states.

Sources:

https://en.wikipedia.org/wiki/Oglala_Lakota_County%2C_South_Dakota

https://en.wikipedia.org/wiki/Union_County%2C_South_Dakota

East Carroll Louisiana: Farming was a main source of employment and the mechanization of agriculture after World War 2 wiped out most of the local economy. Moreover, the population shrank from 19000 to under 10000 citizens which deprived East Carroll of a lot of potential. The region still suffers from racial power structures. Post slavery, many black people stayed in the area and became sharecroppers while the minority white still controls most of the wealth.

Source: <https://time.com/archive/6725828/the-poorest-place-in-america/>

The counties with the lowest poverty rates seem to be remote places in Texas and Utah with very low population sizes. It could either be that wealthy families live there that bring the income per capita up, or that there was no low income reported so the data wouldn't be representative. Loving, Texas is a top producer of oil and gas which generates millions in revenue. Only 8% have an annual revenue of more than \$200,000 a year whereas 60% of the population earn between \$50,000 and \$100,000 a year. So the few wealthy families make the average income higher.

Source:

https://censusreporter.org/profiles/05000US48301-loving-county-tx/?utm_source=chatgpt.com